

Stock Market Prediction Using LSTM Networks

Advanced Financial Forecasting with Deep Learning

Executive Summary



Objectives

Implement a Long Short-Term Memory (LSTM) network to predict sequential price movements with high accuracy.



Data Scope

Utilization of daily historical data from Yahoo Finance for major indices including NVIDIA, Apple, and Tesla.



Key Result

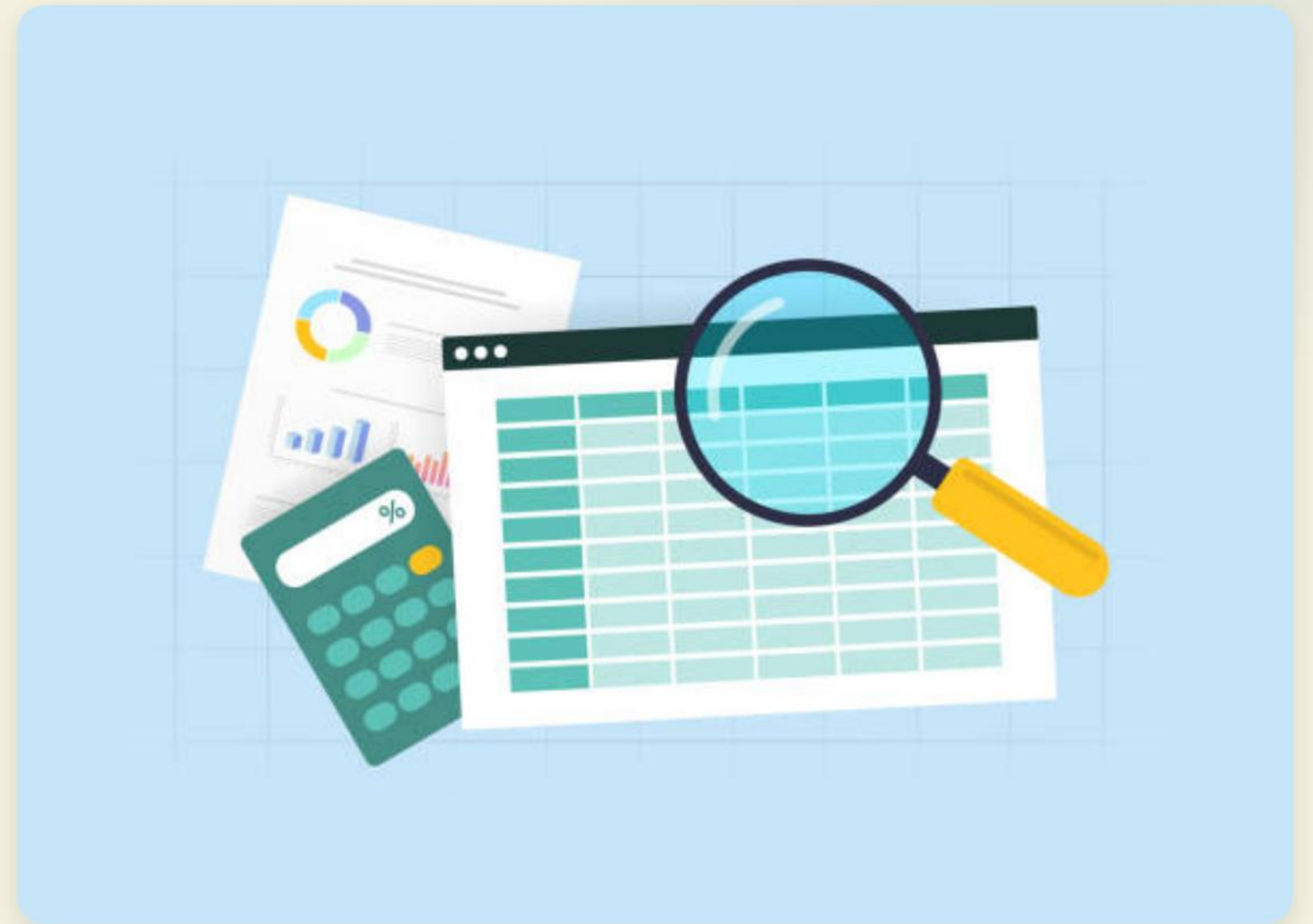
Achieved an R^2 Score of 0.9839 on NVIDIA test data, demonstrating robust trend tracking capabilities.

The Data Challenge

Yahoo Finance Dataset

The model leverages the "Yahoo Finance All Stocks" dataset, accessed via the kagglehub library for seamless integration.

- High-resolution daily closing prices
- Automatic download of latest CSV partitions
- Feature-rich: Open, High, Low, Close, Vol

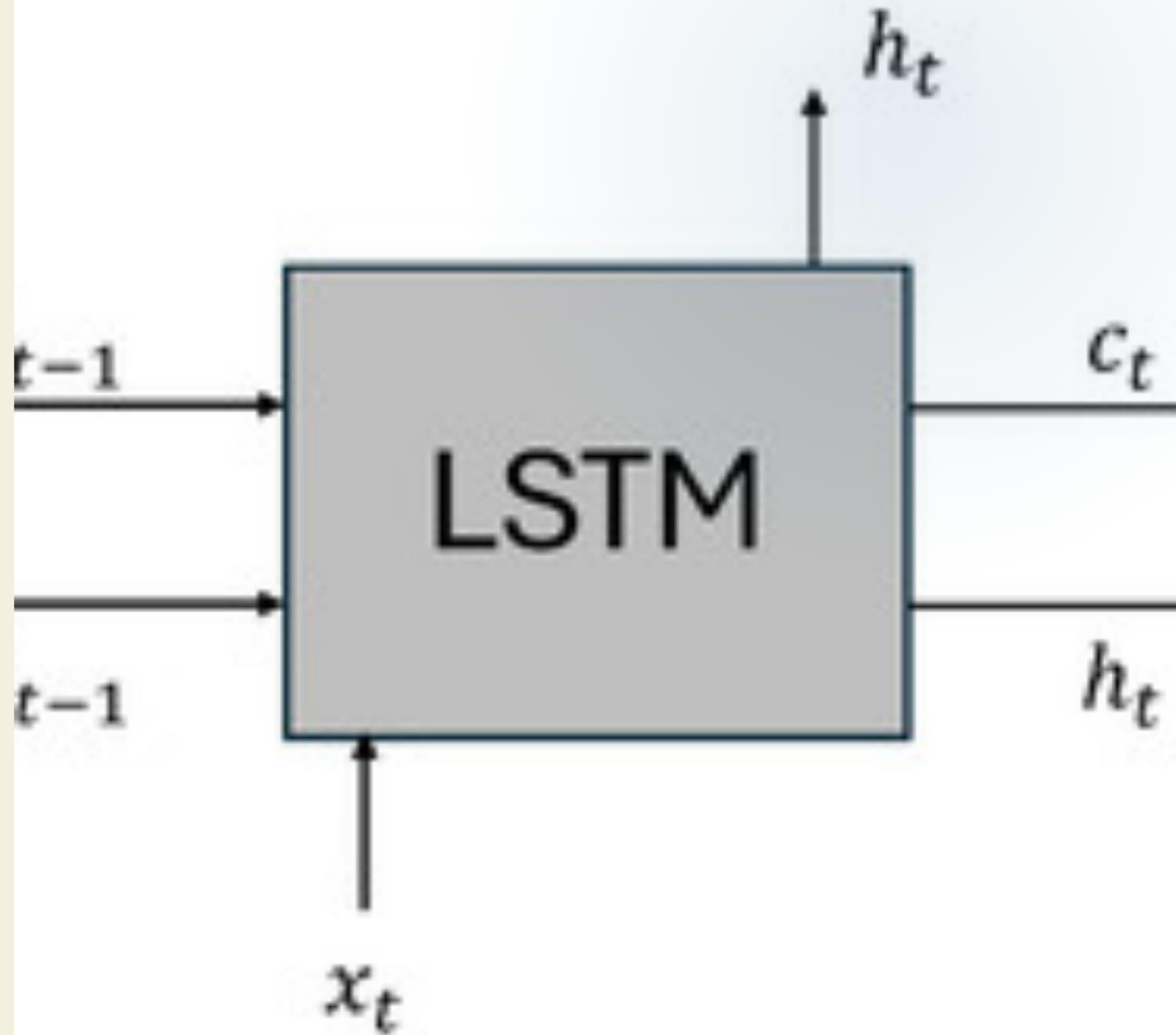


Algorithm Choice

Why LSTM?

Standard Regression models fail to capture "Memory."
Long Short-Term Memory networks are designed for time-series sequences.

LSTM solves the vanishing gradient problem of traditional RNNs, allowing the model to remember historical trends over 60-day windows.



Data Preprocessing

Step	Operation
Cleaning	Removal of NULL values and sorting by Date.
Scaling	Normalizing price to (0, 1) using MinMaxScaler.
Sequencing	Generating 60-day sliding window blocks.
Splitting	80/20 Chronological Train-Test split.

Crucial Step: Feature Scaling

Neural networks converge significantly faster when inputs are scaled. We utilize the formula:

$$X_{\text{scaled}} = \frac{X - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}}$$

Model Construction



Stacked LSTM

Two layers of LSTM cells with 60 units each to extract complex sequential patterns from the price data.



Dropout (0.2)

Regularization technique applied after each layer to prevent overfitting by randomly ignoring neurons during training.



Dense Output

A final fully connected layer that outputs a single continuous value representing the predicted next-day price.

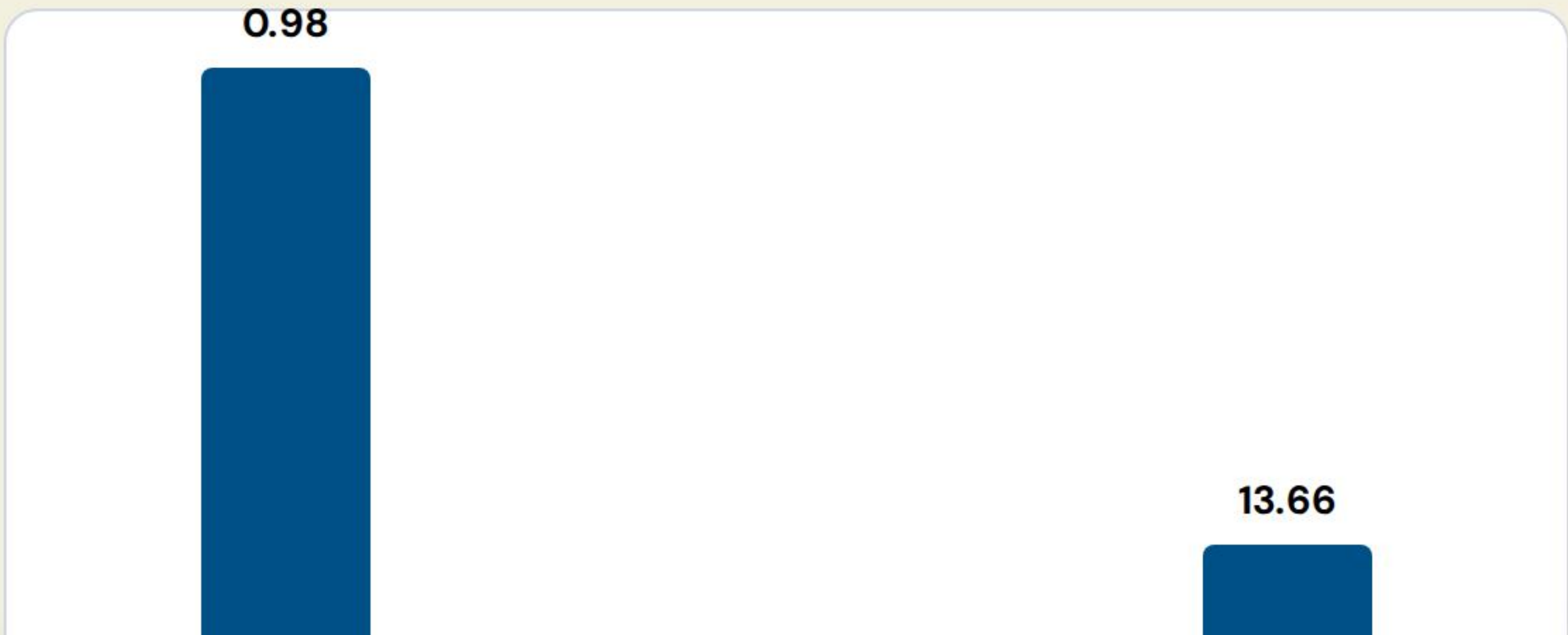
Optimization Strategy

Training Parameters

- ✓ **Optimizer:** Adam (Adaptive Moment Estimation)
- ✓ **Loss Function:** Mean Squared Error (MSE)
- ✓ **Batch Size:** 32
- ✓ **Callback:** EarlyStopping (patience=5)

Adam Optimizer is chosen for its efficiency in handling sparse gradients and non-stationary objectives, making it ideal for the volatile nature of stock market time-series data.

NVDA Performance Metrics



Visualizing Accuracy



Predicted vs. Actual Trend

During the validation phase, the LSTM forecast output (red dashed line) tightly overlaps the Ground Truth actual price (blue solid line). This confirms that the 60-day lookback window is highly predictive of short-term movements.

— Actual Price

--- LSTM Forecast

Tomorrow's Forecast

NVIDIA (NVDA)

\$425.37

Predicted Next Close

APPLE (AAPL)

\$172.81

Predicted Next Close

TESLA (TSLA)

\$267.96

Predicted Next Close

*Predictions based on the absolute latest 60-day sequence from Yahoo Finance daily update.

Conclusion

Project Outcomes

The stacked LSTM model demonstrated superior performance in capturing financial time-series trends.

- Excellent generalization on test data
- Real-world applicability for trading insights
- High precision with R^2 scores above 0.98

Questions?

Stock Market Price Prediction Project

Deep Learning with LSTM Pipeline

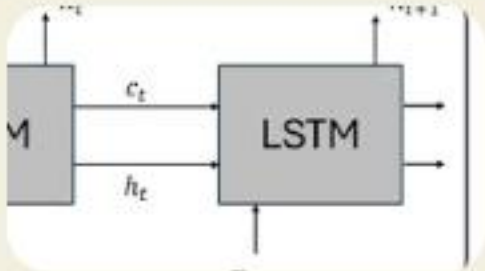
Technical Stack: Python | TensorFlow | Keras | Yahoo Finance

Image Sources



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